

B Analysis of the Impact of Learning Rates and Embedding Dimensions on Model Performance

To provide direct evidence for the use of fixed learning rates and embedding dimensions in MDPRec and baseline models, we selected models encompassing Transformer, MLP, Mamba architectures, frequency modelling, and MDPRec methods for experiments. Specifically, on the LastFM and Video datasets, we fixed all other training parameters and varied only the learning rate or embedding dimension in each iteration to obtain recommendation results from the five models. As shown in Figure 11-12, we observe that: (1) The five models with different architectures exhibit consistent trends in learning rate on the LastFM and Video datasets, and the best results are obtained when the learning rate is 0.001. (2) Except for MDPRec, SASRec, FMLPRec, BSARec exhibit consistent trends in embedding dimension, MDPRec may demonstrate a stronger advantage in high-dimensional modeling due to its unique structure. The Figure 11-12 show that when the embedding dimension is set to 64, four models deliver the best performance, Mamba4Rec delivers competitive performance. Therefore, it is reasonable to set the learning rate to 0.001 and the embedding dimension to 64 for MDPRec and the baseline models; this is also consistent with the settings in the paper [2, 8, 19, 25, 26, 28, 57].

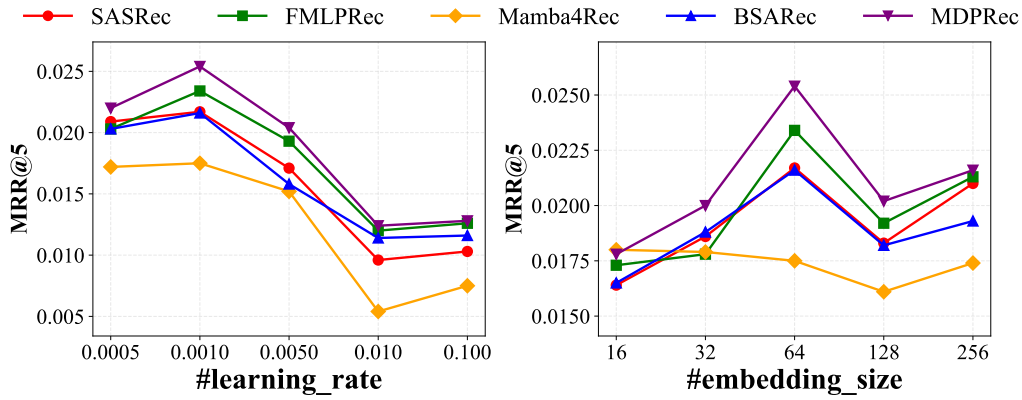


Fig. 11. The impact of different learning rates and embedding dimensions on the performance of various architectural models on the LastFM dataset.

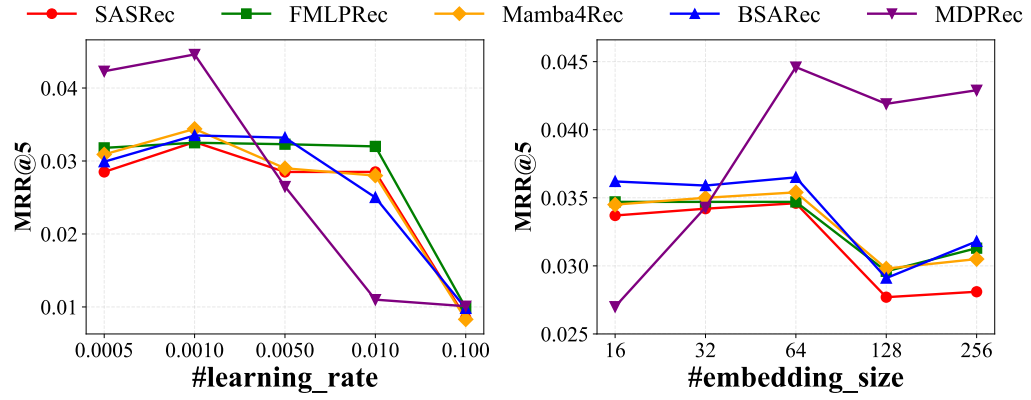


Fig. 12. The impact of different learning rates and embedding dimensions on the performance of various architectural models on the Video dataset.